

Part XII. Compare the Documented and Artificial Mutations

	WT Sequence	Documented mutation	Artificial mutation
CDS Length	1, 386 bp	1, 386 bp	1, 385 bp
Protein Length	461 aa	461 aa	Shorter/abnormal due to premature stop
Mutation Type	None	Missense	Frameshift
Reading frame changed	No	No	Yes
Premature stop codon appeared	No	No	Yes
Amino acids affected	None	1 amino acid (position 226)	Many amino acids from around position 24 onward.
Expected functional consequence	Normal factor IX protein	May alter factor IX function	Likely strongly affects factor IX because the frameshift changes the downstream amino-acid sequence and causes an early stop

The WT F9 sequence has 1,386 bp and produces a normal 461-aa factor IX protein. The documented mutation is a missense mutation that changes only one amino acid at position 226 without changing the reading frame or protein length. In contrast, the artificial mutation deletes one nucleotide, causing a frameshift, changing many downstream amino acids and producing a premature stop codon. Therefore, the artificial mutation is expected to have a stronger effect on the factor IX protein.